

CENG381 Stochastic Processes

Queuing Theory: M/M/1 and M/M/inf — Practice 1

Q1 (30 points). Patients arrive at a clinic according to a Poisson process at rate $\lambda = 4$ per hour. There is a single doctor who serves patients at rate $\mu = 6$ per hour (exponential service times). Model this as an M/M/1 queue.

- Check the stability condition. Compute the traffic intensity $\rho = \lambda/\mu$.
 - For a stable M/M/1 queue the stationary distribution is $\pi_n = (1-\rho) \cdot \rho^n$. Compute π_0 (probability no patients), π_1 , and $P(2 \text{ or more patients in system})$.
 - Compute $E[N] = \rho/(1-\rho)$ (expected number in system) and $E[T] = 1/(\mu-\lambda)$ (expected time in system, waiting plus service). Verify Little's Law: $E[N] = \lambda \cdot E[T]$.
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Q2 (35 points). A car-wash facility has an unlimited number of bays (infinite servers). Cars arrive at rate $\lambda = 6$ per hour (Poisson). Each wash takes $\text{Exp}(\mu)$ time with mean $1/\mu = 20$ minutes = $1/3$ hour, so $\mu = 3$ per hour. Model this as an M/M/ ∞ queue.

- Explain why M/M/ ∞ is always stable regardless of λ and μ . Write the birth-death rate structure: birth rate in state $n = \lambda$ (arrival), death rate in state $n = n \cdot \mu$ (n cars being washed simultaneously).
 - Using detailed balance $\lambda \cdot \pi_n = (n+1) \cdot \mu \cdot \pi_{n+1}$, show that $\pi_n = \pi_0 \cdot \rho^n / n!$ where $\rho = \lambda/\mu$. Enforce $\sum \pi_n = 1$ to find $\pi_0 = e^{-\rho}$. State the distribution of N and compute $E[N]$.
 - With $\lambda=6$ and $\mu=3$, compute ρ , π_0 , π_1 , π_2 , and $P(3 \text{ or more cars being washed})$.
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Q3 (35 points). A network router processes packets at service rate $\mu = 100$ packets/second. Packets arrive as a Poisson process. Assume M/M/1 queuing.

- When the arrival rate is $\lambda = 80$ packets/second, compute: ρ , π_0 , $E[N]$, the expected time in the system $E[T]$, and the expected waiting time in the queue $E[W] = E[T] - 1/\mu$.
- The network SLA requires an average queue waiting time $E[W] \leq 5 \text{ ms} = 0.005 \text{ seconds}$. Using $E[W] = \lambda / [\mu(\mu-\lambda)]$, find the maximum arrival rate λ^* that satisfies this SLA.
- Verify Little's Law $E[N] = \lambda \cdot E[T]$ for the case $\lambda=80$, $\mu=100$. Also: if a second identical router is added (M/M/2), qualitatively describe how $E[W]$ changes compared to M/M/1.